

Hamilton depression rating scale: Uses and applications

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List of abbreviations

HDRS, HRSD, HAM-D	Hamilton depression rating scale
SIGH-D	structured guide for the HDRS
TCAs	tricyclic antidepressants
SSRIs	selective reuptake inhibitors

Introduction

The Hamilton Depression Rating Scale, also known as the Hamilton Rating Scale for Depression (often abbreviated to HDRS, HRSD, or HAM-D), was originally designed to measure the severity of depression and to evaluate the performance of the first group of antidepressants.

In 1960, Max Hamilton, a psychiatrist from the department of psychiatry at Leeds University, presented a rating scale for depression to the psychiatric society (Hamilton, 1960). This scale quickly gained great notability and became widely used within the medical community to assess both the severity of depression in dynamics and the effectiveness of treatment of patients with depressive disorder (Braund, Tillman, Palmer, & Harris, 2020; Cerou, Peigne, Comets, & Chenel, 2020; Liu et al., 2020). It was subsequently revised in 1966 (Hamilton, 1966), 1967 (Hamilton, 1967), 1969 (Hamilton, 1969), and 1980 (Hamilton, 1980). Over the past 50 years, the HAM-D has been modified into 11 versions and applied in various patient populations in psychiatric, medical, and other scientific settings (Rohan et al., 2016; Williams, 2001). Still, the HDRS is widely available in the public domain and is not protected by copyright.

Attempts were made to modify the scale, both by including more signs of depression and by increasing the degree of gradation of each of item, a more detailed assessment of their intensity or frequency in different revisions of scale. For more than 40 years it has been the gold standard for the assessment of depression and serves as a

point of reference for more recently developed scales (Table 1). However, criticism of the instrument has been increasing (Bagby, Ryder, Schuller, & Marshall, 2005).

Administration and uses

The HDRS is an interviewer-rated scale for adults with already identified depression. The HADS scale should not be used as a diagnostic instrument. It requires a trained rater with knowledge of the tool and symptoms of the depressive illness. The HADS is designed to be used by a health care professional during a clinical interview and to measure the outcome of treatment, especially drugs. Hamilton indicated that the value of the standard HADS scale depends on the skills of the interviewer (Hamilton, 1960).

In 1988, the Structured Guide for the HDRS (SI-HDRS) was developed to standardize the manner of the administration of the scale (Williams, 1988). A study conducted on a series of psychiatric inpatients demonstrated a better level of agreement for most of the HDRS items. Not only experienced clinicians can apply the SI-HDRS items, but also trained interviewers who lack psychiatric background. In addition, some studies have pointed out that a structured interview does not differ between face-to-face and telephone administration groups. Thus, this instrument can be used economically in large-scale, community-based research projects (Potts, Daniels, Burnam, & Wells, 1990).

Through repeated and consistent use of the scale, a clinician can document the results of treatment. The assessment typically takes 15–30 min and can be used both for inpatients and for patients outside a psychiatric clinical setting.

Scoring and interpretation

The original version contained 17 items, but during the last 5 decades, the researchers have modified it for certain

TABLE 1 Rating scales for depression.

Scale	Acronym	Number of items	Administration time	Indication
<i>Scales completed by professionals</i>				
Composite International Diagnostic Interview ^a	CIDI	41 sections of items	20–30 min	Diagnostic tool based on DSM-IV
Hamilton Depression Rating Scale ^b	HDRS HAM-D HRSD	17–21	15–30 min	Severity of depression Treatment outcomes
Montgomery-Asberg Depression Rating Scale ^c	MADRS	10	20–40 min	Severity of depression Treatment outcomes
Structured Clinical Interview for DSM ^d	SCID-I SCID-II SCID-5	Depend on version	15 min–2 h	Diagnostic tool based on DSM-IV or DSM-5
Symptom Checklist 90-Revision ^e	SCL-90-R	90	12–15 min	Severity of symptoms Treatment outcomes
The Mini International Neuropsychiatric Interview ^f	M.I.N.I.	17 modules of items	15 min	Diagnostic tool based on DSM-III-R, DSM-IV, DSM-5 and ICD-10
<i>Scales completed by patient, self-rating scales</i>				
Beck Depression Inventory ^g	BDI	21	5–10 min	Screening tool Severity of DS Treatment outcomes
Center of Epidemiologic Studies—depression ^h	CES-D	20	15 min	Screening tool
Geriatric Depression Scale ⁱ	GDS-4 GDS-15 GDS-30	4 15 30	3 min 5–7 min 10–20 min	Screening tool Severity of DS Treatment outcomes
Hospital Anxiety and Depression scale ^j	HADS	14	2–5 min	Screening tool Severity of symptoms Treatment outcomes
Inventory of depression and Anxiety Symptoms ^k	IDAS	99	20–30 min	Measures symptoms of depression and anxiety
Patient Health Questionnaire for depression—9 ^l	PHQ-9	9	3 min	Screening tool Severity of DS Treatment outcomes
Self-Report Questionnaire—20 ^m	SRQ-20	20	10–15 min	Screening tool Treatment outcomes
WHO Well-Being Index ⁿ	WHO-5	5	3 min	Screening tool Treatment outcomes

TABLE 1 Rating scales for depression—cont'd

Scale	Acronym	Number of items	Administration time	Indication
Zung's Self-Rating Depression Scale ^o	SDS ZSDS	20	10–20 min	Screening tool Severity of DS Treatment outcomes
<i>Scales completed by professionals and patients</i>				
The Primary Care Evaluation of Mental Disorders ^p	PRIME-MD	27 and clinician interview	10 min	Screening tool Diagnostic tool

^aKessler and Ustun (2004).^bHamilton (1960, 1966, 1967, 1969, 1980).^cMontgomery and Åsberg (1979).^dShankman et al. (2018).^eDerogatis (1977).^fSheehan et al. (1998).^gBeck, Ward, Mendelson, Mock, and Erbaugh (1961).^hRadloff (1977).ⁱYesavage et al. (1982).^jZigmond and Snaith (1983).^kWatson et al. (2007).^lSpitzer, Kroenke, and Williams (1999).^mBeusenbergh et al. (1994).ⁿTopp, Østergaard, Søndergaard, and Bech (2015).^oZung (1965).^pSpitzer et al. (1994).

Rating scales of depression are comprised of three groups: scales completed by professionals, completed by patients (self-rating scales) and by both. The administration time varies from 3 min to 2 h.

DS, depressive symptoms; *DSM-IV*, the Diagnostic and Statistical Manual of Mental Disorders, 4th Edition; *DSM-V*, 5th Edition; *DSM-III-R*, 3rd Edition; *ICD-10*, International Classification of Diseases 10th Edition. Data are from

needs. Subsequently, attempts were made to expand or shorter the scale, both by cutting-off or including more items, ranging from 6 to 36, or transforming the assessment into a more detailed one concerning changes in intensity or frequency (McIntyre, Kennedy, Bagby, & Bakish, 2002; Willaims, 2001).

For example, a shortened form with seven items that are identified and designated as the Toronto HAM-D7 with cutoff scores for full remission offers a useful tool that physicians can readily employ in clinical practice (McIntyre et al., 2002). Longer versions were developed to cover reverse neurovegetative, atypical symptoms of depression (Willaims, 2001). It should be noted that special studies showed little success in expanding the list of features included in the HDRS since this did not enhance the reliability and validity of this scale.

However, the most commonly used versions in the studies are either a 17- or a 21-item scale.

The first 17 items measure the severity of depressive symptoms experienced over the past week and include depressed mood, guilt, suicide, initial insomnia, middle insomnia, delayed insomnia, work and interests, retardation, agitation, psychic anxiety, somatic anxiety, gastrointestinal symptoms, general somatic symptoms, genital symptoms, hypochondriasis, loss of insight and weight (Table 2).

A 21-item version (HDRS-21) includes additional 4 items (18–21 items) intended to identify a subtype of depression and to measure factors that might be related to depression, but are not thought to be measures of severity, such as paranoia, obsessional symptoms, depersonalization and derealization, and diurnal variation (Table 2). These points are not taken into account when calculating the total score of the Hamilton scale, which determines the severity of depressive disorder, thus decreasing the internal consistency as a whole.

TABLE 2 Hamilton depression rating scale: Symptoms and score range.

Item no.	Symptom	Score range
1.	Depressed mood	0–4
2.	Guilt	0–4
3.	Suicide	0–4
4.	Initial insomnia	0–2
5.	Middle insomnia	0–2
6.	Delayed insomnia	0–2
7.	Work and interests	0–4
8.	Retardation	0–4
9.	Agitation	0–2
10.	Psychic anxiety	0–4
11.	Somatic anxiety	0–4
12.	Gastrointestinal symptoms	0–2
13.	General somatic symptoms	0–2
14.	Genital symptoms	0–2
15.	Hypochondriasis	0–4
16.	Loss of insight	0–2
17.	Loss of weight	0–2
18.	Diurnal variation	0–2
19.	Depersonalization	0–4
20.	Paranoid symptoms	0–4
21.	Obsessional symptoms	0–2

Data are from Hamilton, M. (1960). A rating scale for depression. *Journal of Neurology, Neurosurgery & Psychiatry*, 23(1), 56. <https://doi.org/10.1136/jnnp.23.1.56>.
The symptoms and score range are shown for the 21-item scale.

TABLE 3 Hamilton depression rating scale: Grading of scores.

	Severity of symptom
<i>Score range 0–4</i>	
0.	Absent
1.	Mild or trivial
2.	Moderate
3.	
4.	Severe
<i>Score range 0–2</i>	
0.	Absent
1.	Slight or doubtful
2.	Clear present

Data are from Hamilton, M. (1960). A rating scale for depression. *Journal of Neurology, Neurosurgery & Psychiatry*, 23(1), 56. <https://doi.org/10.1136/jnnp.23.1.56>.

The symptoms severity are evaluated on a five-point scale (0–4) and on a three-point scale (0–2).

Depending on the item, it is scored between 0 (not present) and 4 (severe) points using either a three-point or a five-point scale and summed up to obtain the total score. The HDRS comprises 17 items, of which 9 are evaluated on a five-point scale (0–4) and 8—on a three-point scale (0–2). The total score can range from 0 to the maximum score 52 on a 17-point scale, with higher scores indicating more serious depression (Tables 2 and 3).

For the HDRS-17, scores less than 7 are indicative of the absence of depression or being normal, 8–16 consider mild depression, 17–23 suggest moderate depression and scores over 24 are indicative of severe depression (Zimmerman, Chelminski, & Posternak, 2004) (Table 4).

A decrease of 50% or more in the HDRS score is often considered as a positive treatment response; on the other hand, a score of 7 or less considered equivalent to a remission (Zimmerman et al., 2004) (Table 5).

Indication for the use of the HDRS

In most clinical trials, the HDRS is used as a standard to determine the severity of depression and response to the treatment. The HDRS is the most common tool for evaluating the effectiveness of antidepressants in clinical trials (Leader, O’Connell, & VandenBerg, 2019). According to the criteria of the National Institute for Clinical Excellence, the effect of an antidepressant compared to placebo is clinically significant, if the difference between the effect of antidepressant and placebo is only three points or more on the Hamilton scale (Andrews, Thomson, Amstadter, & Neale,

TABLE 4 Hamilton depression rating scale 17-item: Interpretation.

Score range	Severity level
7 or <	Absence of depression or being normal
8–16	Mild depression
17–23	Moderate depression
24 or >	Severe depression

The total score range from 0 to the maximum score 52

Data are from Zimmerman, M., Martinez, J. H., Young, D., Chelminski, I., & Dalrymple, K. (2013). Severity classification on the Hamilton depression rating scale. *Journal of Affective Disorders*, 150(2), 384–388. <https://doi.org/10.1016/j.jad.2013.04.028>.

The scores less than 7 or 7 are indicative of the absence of depression or being normal, 8–16 consider mild depression, 17–23 suggest moderate depression and scores 24 or over are indicative of severe depression.

TABLE 5 Hamilton depression rating scale 17-item: Response to treatment.

- A decrease of 50% or more in the HDRS score is often considered as a positive treatment response
- A score of 7 or less considered equivalent to a remission

Data are from Zimmerman, M., Chelminski, I., & Posternak, M. (2004). A review of studies of the Hamilton depression rating scale in healthy controls: Implications for the definition of remission in treatment studies of depression. *The Journal of Nervous and Mental Disease*, 192, 595–601.

^a Zimmerman, M., Chelminski, I. & Posternak, M. 2004. A Review of Studies of the Hamilton Depression Rating Scale in Healthy Controls: Implications for the Definition of Remission in Treatment Studies of Depression. *The Journal of Nervous and Mental Disease*, 192, 595–601. The interpretation of changes in the scores during the treatment is shown in Table 5.

2012). In other studies, evaluating the level of depression, the HDRS is used to assess the effectiveness of psychotropic drugs of different classes, but not only antidepressants (An et al., 2019; Fayed et al., 2019; Kimball et al., 2019).

In addition, there are many studies evaluating the effectiveness of non-pharmacological interventions on depression. It was used to measure the effectiveness of psychotherapy, neurostimulation, complementary, and alternative treatment of depression (chronotherapy, light therapy, physical exercises, and others) (An et al., 2019; Arslanoglou, Banerjee, Pantelides, Evans, & Kiosses, 2019; Elboga, Karayagmurlu, Kocamer Sahin, & Altindag, 2019; Erbay, Zayman, Erbay, & Unal, 2019; Wang et al., 2019; Yrondi et al., 2019). In several studies, the HDRS was used as a tool to measure suicidality (Shen et al., 2019; Williams et al., 2019).

Although Hamilton recommended to use the scale only with patients already diagnosed affective disorder of the depressive type, many researchers use a scale to measure

the level of depression for patients suffering from psychotic disorders, alcohol and drug addictions, neurotic disorders and other mental disorders (Kimball et al., 2019; Saglam, Aslan, & Hursitoglu, 2020; Tonna et al., 2019).

Scientists used this scale to measure the level of depression in patients suffering from general medical conditions, neurological or neurodegenerative diseases (Che et al., 2019; Chen et al., 2019; Fayed et al., 2019; Hassaan et al., 2019; Shen et al., 2019).

The HDRS can be used in place of self-report scales, when a patient is unable to read or in case when there are concerns about the accuracy of the patient's self-report (Halfaker, Akeson, Hathcock, Mattson, & Wunderlich, 2011).

This scale can be useful for patients with cognitive impairment who experience difficulty managing self-reporting tools; and the scale has demonstrated reliability, validity, and efficiency in older adults above 60 years (Arslanoglou et al., 2019; Tulaci & Ekinci, 2020) (Table 6).

TABLE 6 Hamilton depression rating scale: Indication for the use.

- To determine the severity of depression and response to treatment^a
- For evaluating the effectiveness of antidepressants^b and others psychotropic medications^{c,d,e}
- For evaluating the effectiveness of nonpharmacological interventions: psychotherapy, neurostimulation, complementary and alternative treatment of depression (chronotherapy, light therapy, physical exercises and others)^{d,f,g,h,i,j}
- To measure suicidality^{k,l}
- To measure the level of depression for patients suffering from psychotic disorders, alcohol and drug addictions, neurotic disorders and other mental disorders^{m,n,o}
- To measure the level of depression in patients suffer from general medical conditions, neurological or neurodegenerative diseases^{c,e,l,m,n,o,p}
- Can be used in place of self-reports scales (if patient unable to read or patients has cognitive impairment)^{c,q}

Data are from

^aHamilton (1960).

^bLeader et al. (2019).

^cFayed et al. (2019).

^dAn et al. (2019).

^eKimball et al. (2019).

^fArslanoglou et al. (2019).

^gElboga et al. (2019).

^hWang et al. (2019).

ⁱYrondi et al. (2019).

^jErbay et al. (2019).

^kWilliams (1988).

^lShen et al. (2019).

^mSaglam et al. (2020).

ⁿTonna et al. (2019), Chen et al. (2019).

^oChe et al. (2019).

^pHassaan et al. (2019).

^qTulaci and Ekinci (2020).

The general indication for the use of scale is shown in Table 6.

Validity and reliability

Since 1960, validation of the scale has been performed in a wide array of studies. The researchers also evaluated consistency of the results obtained in different studies. The reviews of recent years demonstrate that the scale has generally satisfactory psychometric characteristics.

Major psychometric properties of the HDRS were examined by Bagby et al. (2005). In the current review of 70 studies, the authors confirmed that the internal, interrater and retest reliability estimates are mostly good for the overall rating score. However, the internal reliability coefficients show that some items are problematic, and internal and retest coefficient are weak for many individual items. The data of the review suggest that the convergent, discriminant, and predictive validity meet established criteria (Bagby et al., 2005).

However, the validity of the HAMD has often been criticized in recent years. Some items of the scale are distinguished by constructive inadequacy, which influences the accuracy of assessing the severity of depression. It is noted that threshold criteria that distinguish a mild degree of depression from remission do not meet criteria of validity. Some of the items exhibit weak construct validity (Romera, Pérez, Menchón, Polavieja, & Gilaberte, 2011; Sawamura, Ishigooka, & Nishimura, 2018).

Limitations

The HDRS was developed before the publication of the Diagnostic and Statistical Manual of Mental Disorders, 4th or 5th Edition and does not evaluate all domains of the diagnostic criteria for Major Depressive Disorder such as feelings of worthlessness and anhedonia (Santor & Coyne, 2001).

The HRSD is criticized for being used in clinical practice as it places more emphasis on insomnia rather than on feelings of hopelessness, self-destructive thoughts, suicidal cognitions and actions (Brown et al., 2000); it also favors somatic signs and symptoms and can miss atypical symptoms, such as overeating and oversleeping. A meta-analysis data suggests that after therapeutic treatments, the HRSD is more 'sensitive to change' on retesting in comparison to the Beck Depression Inventory (self-rating instrument) and this is probably why it has been so widely used in clinical trials (Brown et al., 2000). In addition, from a critical point of view, in case when suicidal thoughts increase and sleep is improved, an antidepressant may show statistical efficacy in a clinical trial. Some studies show that the HDRS has priority sedative tricyclic antidepressants (TCAs) above selective reuptake inhibitors (SSRIs) (Möller, 2001; Santor and Coyne, 2001; Worboys, 2012). The side effect of TCAs could influence the sleep and somatic items; and such side effects as insomnia, agitation, gastrointestinal and sexual side effect of SSRIs could be rated on a scale and may be registered as being less effective

TABLE 7 Hamilton depression rating scale (HRSD): Limitations.

- It does not evaluate all domains of the diagnostic criteria for Major Depressive Disorder in DSM-IV or DSM-V such as feelings of worthlessness and anhedonia^a
- It places more emphasis on insomnia rather than on feelings of hopelessness, self-destructive thoughts, suicidal cognitions and actions^b
- It favors somatic signs and symptoms and can miss atypical symptoms, such as overeating and oversleeping^b
- After therapeutic treatments, the HRSD is more “sensitive to change” on retesting in comparison to the Beck Depression Inventory^b
- In case when suicidal thoughts increase and sleep is improved, an antidepressant may show statistical efficacy
- The side effect of TCAs could influence the sleep and somatic items^{a,c,d}
- The side effects of SSRIs as insomnia, agitation, gastrointestinal and sexual side effect of SSRIs could be rated on scale^{c,e,f}
- Atypical symptoms of depression such as hypersomnia and hyperphagia are not assessed

Data are from

^aSantor and Coyne (2001).

^bBrown et al. (2000).

^cMöller (2001).

^dWorboys (2012).

^eBagby et al. (2005).

^fLinden et al. (1995).

The limitation of the scale are mentioned in Table 7.

DSM-IV, the Diagnostic and Statistical Manual of Mental Disorders, 4th Edition; DSM-V, the Diagnostic and Statistical Manual of Mental Disorders, 5th Edition; TCAs, tricyclic antidepressants; SSRIs, selective reuptake inhibitors.

in treating the depression than it actually is (Bagby et al., 2005; Linden, Borchelt, Barnow, & Geiselmann, 1995; Möller, 2001). Another limitation of the HDRS-17 is that atypical symptoms of depression such as hypersomnia and hyperphagia are not assessed (Table 7).

Key facts of the Hamilton depression rating scale

- The Hamilton Depression Rating Scale (HDRS) was originally designed to measure the severity of depression symptoms and response to treatment.
- The HDRS is an interviewer-rated scale for adults with already identified depression.
- The HADS scale should not be used as a diagnostic instrument.
- In 1988, the Structured Guide for the HDRS was developed to standardize the manner of the administration of the scale.

- The most commonly used versions of the HDRS in the studies are either a 17- or a 21-item scale.
- The first 17 items measure the severity of depressive symptoms experienced over the past week.
- A 21-item version includes additional four items (18–21 items) intended to identify a subtype of depression. These points are not taken into account when calculating the total score of the Hamilton scale.
- The HDRS does not evaluate all domains of the diagnostic criteria for Major Depressive Disorder. The side effects of antidepressants could influence the score of items.

Summary points

- This chapter focuses on the Hamilton Depression Rating Scale which is a widely used measure instrument to assess the severity of depression in dynamics and the effectiveness of treatment of patients with already identified depressive disorder.
- A trained rater with knowledge of the tool and symptoms of the depressive illness should administer it.
- The most commonly used versions in the studies are either a 17- or a 21-item scale.
- The scoring of the severity of the depressive symptoms is based on the first 17 items. It is scored between 0 (not present) and 4 (severe) points using either a three-point or a five-point scale and summed up to obtain the total score.
- The assessment generally takes 15–30 min.
- The scale does not evaluate all domains of the diagnostic criteria for Major Depressive Disorder and the side effects of antidepressants could influence the score of items.

Mini-dictionary of terms

Depression rating scale A measuring instrument intended for a comparative assessment of the severity of depressive symptoms in dynamics in cases where clinical diagnosis is already established.

Structured guide for rating scale Standardized way of interviewing with a series of the same questions in the same order, designed to assess the severity of symptoms of the scale.

Treatment response. A rating scale score changes during a continuous measure to evaluate the treatment outcome (response, remission, nonresponse, partial response, relapse, recurrence, recovery, and depressive breakthrough).

Positive treatment response A decrease of 50% or more in the initial rating scale scores.

Nonresponse to treatment Changes below 50% in the initial rating scale scores.

Remission A score of the Hamilton Depression Rating Scale 7 or less.

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